

Juliet Cohen

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SUMMARY OF QUALIFICATIONS

- Develops and deploys reproducible workflows in Python to process large, remotely sensed datasets on UC Santa Barbara's high performance computing clusters
- Executed statistical and geospatial analysis in R for the Ocean Health Index at the National Center for Ecological Analysis and Synthesis
- Applied machine learning methods to process satellite imagery in Python and develop a predictive model
- Collaboratively manages multi-terabyte databases and metadata at the Arctic Data Center

EDUCATION

Master of Environmental Data Science 4.0 GPA (June 2022)

Bren School of Environmental Science & Management - University of California, Santa Barbara

Highlighted Coursework: Remote Sensing and Environmental Data; Spatial Analysis for Environmental Data Science

Bachelor of Science in Ecology and Evolution

3.7 GPA (June 2019)

University of California, Santa Barbara

Honors: Distinction in the Major

Study Abroad: Monteverde Institute, Costa Rica - Tropical Biology and Conservation Program

Athletics: Rowing Team

MASTER'S CAPSTONE PROJECT

An open-source pipeline for remote sensing of crop yields: a Zambia case study

(1/22 – 6/22)

Role: Data Manager | Clients: Dr. Tamma Carleton, Dr. Jonathan Proctor

- Developed an open-source tool in Python to process satellite imagery for modeling environmental trends through both unsupervised and supervised machine learning (see [project organization](#) on GitHub)
- Contributed to a task-agnostic tool for researchers to monitor the impact of climate change and socioeconomic factors over time and space
- Presented master's project and its environmental justice implications for the *Justice, Equity, Diversity, and Sustainability Initiative* at the New Horizons in Conservation Conference at the Yale School of the Environment (see [Programming Blog](#) on website)
- Executed statistical analysis, documented metadata, and contributed to writing a scientific paper

ENVIRONMENTAL DATA SCIENCE WORK EXPERIENCE

Arctic Data Center, National Center for Ecological Analysis and Synthesis (NCEAS)

Data Scientist & Software Developer

(10/22 – present)

- Processes large datasets derived from satellite imagery and machine learning models in Python
- Develops and deploys containerized workflows on UCSB's high performance computing clusters with parallelization
- Visualizes datasets on the [Permafrost Discovery Gateway](#), an interactive data portal ([GitHub organization](#))
- Collaboratively manages multi-terabyte databases and metadata to increase accessibility and reproducibility
- Engages in community outreach through presentations, conferences, and workshops
- Participates in the Diversity, Equity, Inclusion, and Justice Seminar Series Planning Committee

Ocean Health Index, NCEAS

Data Scientist Fellow

(5/22 – 9/22)

- Synthesized global datasets related to wildlife, anthropogenic impacts, and human well-being
- Statistically calculated future trajectories of biodiversity, industrial fishing in exclusive economic zones, etc.
- Communicated scientific results through interactive visualizations, maps, open-source code, documentation, and technical [blog posts](#)

-Continued-

ENVIRONMENTAL DATA SCIENCE WORK EXPERIENCE - Continued

Arctic Data Center, NCEAS

Data Intern

(1/22 – 4/22)

- Database curation in R, managed Arctic datasets and metadata with reproducible workflows
- Associated related data with semantic annotations, categorization, and provenance

CONSERVATION WORK EXPERIENCE

Pacific States Marine Fisheries Commission & CDFW - Fisheries Technician

(12/20 – 6/21)

- Monitored endangered Southern California steelhead trout populations in human-occupied regions

Oahu Invasive Species Committee - Data Specialist & Field Technician

(9/19 – 8/20)

- Served as data specialist and crew leader in field surveys for incipient invasive flora and fauna on Oahu

San Diego Natural History Museum - Field Technician

(5/19 – 7/19)

- Led field surveys of the flat-tailed horned lizard and handled reptiles

UC Santa Barbara McCauley Lab - Research Assistant

(9/17 – 5/19)

- Conducted mesocarnivore spatial ecology research studying anthropogenic impacts on behavior

Cheadle Center for Biodiversity and Ecological Restoration – Field Technician

(1/18 – 5/19)

- Wetland field restoration, herbarium maintenance, and botanical specimen preparation

PUBLISHED DATASETS & ECOLOGICAL RESEARCH (see [Research](#) on website)

Annett Bartsch, Georg Pointner, Ingmar Nitze, Juliet Cohen, Barbara Widhalm, Clemens von Baeckmann, & Rodrigue Tanguy. (2024). *Arctic Infrastructure: Sentinel-1 and Sentinel-2 derived Arctic Coastal Human Impact dataset, Pan-Arctic Region, 2016 - 2020*. Arctic Data Center. [doi:10.18739/A21J97929](https://doi.org/10.18739/A21J97929).

Chandi Witharana, Mahendra R Udawalpola, Amal S Perera, Amit Hasan, Elias Manos, Anna Liljedahl, Mikhail Kanevskiy, M Torre Jorgenson, Ronald Daanen, Benjamin Jones, Howard Epstein, Matthew B Jones, Robyn Thiessen-bock, Juliet Cohen, & Kastan Day. (2023). *Ice-wedge polygon detection in satellite imagery from pan-Arctic regions, Permafrost Discovery Gateway, 2001-2021*. Arctic Data Center. [doi:10.18739/A2KW57K57](https://doi.org/10.18739/A2KW57K57).

Ingmar Nitze, & Juliet Cohen. (2024). *Pan-Arctic lake area time series (2017-2021)*. Arctic Data Center. (in progress)

S. Jeanette Clark, Matthew B. Jones, Samantha Csik, Carmen Galaz García, Bryce Mecum, Natasha Haycock-Chavez, Daphne Virlar-Knight, Juliet Cohen, Anna Liljedahl. 2023. *Scalable and Computationally Reproducible Approaches to Arctic Research*. Arctic Data Center. [doi:10.18739/A2QF8JM2V](https://doi.org/10.18739/A2QF8JM2V)

[Monitoring Maize Yield Variability over Space and Time with Unsupervised Satellite Imagery Features](#)
(in progress)

[Anthropogenic niche partitioning: mesocarnivore spatial and temporal coexistence along an urban gradient through camera traps](#)

[Filtration Efficiency in Bivalves: effects of species and size in oysters and mussels](#)

AWARDS & SCHOLARSHIPS

Undergraduate Research and Creative Activities Grant | \$750 | UC Santa Barbara (2018)

Dean Bazzi Memorial Scholarship | \$500 | UC Santa Barbara (2018)

UC Santa Barbara EAP Gaucho Scholarship | \$2,000 | UC Santa Barbara (2018)

SKILLS

Programming Languages: R, Python, SQL, bash, Slurm Workload Manager for workflow deployment

Software Proficiency: ArcGIS, Microsoft Access, Microsoft Suite, DIDSON sonar metrics